Name: Dwijesh Mistry
Class / Batch: TY4 / B
Roll no: 38

EXPERIMENT NO – 7

Aim: Implementation of Data Discretization (any one) and Data Visualization (any one)

Course Outcome: Perceive the importance of data pre-processing and basics of data mining techniques.

Requirement: IDLE

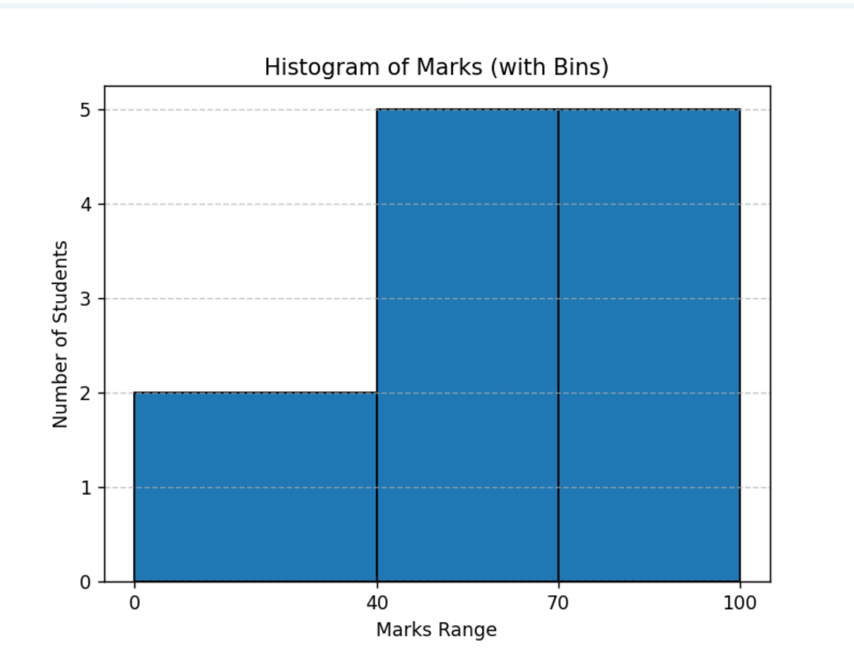
Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Create sample data
data = {
    'Marks': [23, 45, 56, 67, 78, 89, 90, 34, 65, 72, 88, 55]
}

df = pd.DataFrame(data)

print("Original Data:")
print(df.to_string(index=False))

# Step 2: Data Discretization (Equal-width binning)
bins = [0, 40, 70, 100]
labels = ['Low (0-40)', 'Medium (40-70)', 'High (70-100)']

df['Category'] = pd.cut(df['Marks'], bins=bins, labels=labels)

print("\nDiscretized Data (Table Format):")
print(df.to_string(index=False))

# Step 3: Show which values fall into each bin
grouped = df.groupby('Category')['Marks'].apply(list)

print("\nValues in Each Bin:")
for category, values in grouped.items():
    print(f"{category}: {values}")

# Step 4: Visualization (Improved Histogram)
plt.hist(df['Marks'], bins=bins, edgecolor='black')
```

```
plt.title("Histogram of Marks (with Bins)")
plt.xlabel("Marks Range")
plt.ylabel("Number of Students")

# Show bin labels on x-axis
plt.xticks(bins)

plt.grid(axis='y', linestyle='--', alpha=0.7)

plt.show()
```

Output:

```
Original Data:
 Marks
    23
    45
    56
    67
    78
    89
    90
    34
    65
    72
    88
    55

Discretized Data (Table Format):
 Marks        Category
    23      Low (0-40)
    45 Medium (40-70)
    56 Medium (40-70)
    67 Medium (40-70)
    78  High (70-100)
    89  High (70-100)
    90  High (70-100)
    34      Low (0-40)
    65 Medium (40-70)
    72  High (70-100)
    88  High (70-100)
    55 Medium (40-70)

Values in Each Bin:
Low (0-40): [23, 34]
Medium (40-70): [45, 56, 67, 65, 55]
High (70-100): [78, 89, 90, 72, 88]
```



Review Questions:

Q1. How does the process of Equal Width Discretization transform continuous data into discrete categories, and what role does the choice of the number of bins play in the outcome?

Answer:
Equal Width Discretization transforms continuous data into discrete categories by dividing the entire data range into intervals (bins) of equal size.

Process:
1. Find minimum and maximum values in the dataset
2. Compute bin width:

$$\text{Bin Width} = \frac{\text{Max} - \text{Min}}{\text{Number of Bins}}$$

3. Divide the range into equal intervals
4. Assign each data value to its corresponding bin

Example:
Marks (0–100) with 3 bins:
- 0–33 → Low
- 34–66 → Medium
- 67–100 → High

Role of Number of Bins:
The number of bins directly affects how data is grouped:
- **Too few bins**:
  - Data becomes over-simplified
  - Important details are lost
- **Too many bins**:
  - Data becomes too fragmented
  - Patterns become harder to identify

So, choosing the right number of bins is important for balanced analysis.

Q2 What insights can be derived from the histogram of the original continuous data, and how does it help in understanding the distribution of the data after discretization?

Answer:
A histogram visually represents the frequency distribution of continuous data.

From the histogram, we can observe:
- Shape of data (normal, skewed, uniform)
- Spread of values
- Concentration of data (where most values lie)
- Presence of outliers

Relation to Discretization:
- Histogram shows how data is naturally distributed

- Helps in deciding appropriate bin ranges for discretization
- After discretization, histogram confirms whether:
  - Data is evenly distributed across bins
  - Some bins are overloaded or empty

Q3 What are the potential advantages and limitations of using Equal Width Discretization for continuous data, and how can the choice of bin width affect the analysis of the data?

Answer:

Advantages:
- Simple and easy to implement
- Fast computation
- Works well for uniformly distributed data
- Easy to interpret

Limitations:
- Sensitive to outliers (extreme values affect bin width)
- May produce uneven distribution of data across bins
- Not suitable for skewed data
- Some bins may have too many or too few values

Effect of Bin Width:
- **Large bin width (fewer bins)**:
  - Loss of detail
  - Oversimplified data
- **Small bin width (more bins)**:
  - More detailed analysis
  - But may introduce noise and complexity